

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1702

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I NEELAM NANDINI ^{C192472277} (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

18
20



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

A smart home automation system.

Introduction

A smart home automation system uses AI to automatically control lighting, temperature and security based on user behavior and environmental conditions. It receives input from sensors such as motion detectors, temperature sensors and time setting. Then decides the most appropriate action. Intelligent agents make these decisions.

Types of Intelligent Agents.

1. Simple Reflex Agent

Definition - A simple Reflex Agent makes decisions only based on the current input using predefined rules. It does not remember previous events.

Working in smart Home

If motion is detected → turn on the lights

If no motion for 10 min → turn off the lights

If door sensors detects intrusion → sound the alarm

Advantages

Fast response

Easy to implement

Low computational cost

Disadvantages

cannot learn user habits

cannot predict future situations

may waste energy if rules are not sufficient.

2. Model Based Reflex Agent

Definition - A model-based Agent maintains an internal model (memory) of the environment and uses past information of decision-making.

Working in smart Home.

Remembers that the bedroom light is already on. (on).

knows whether doors are locked.

Tracks room temperature over time.

If temperature keeps increasing it starts cooling before the room becomes uncomfortable.

Advantages

- Handles partially observable environments
- Makes better decisions using memory
- More reliable than simple reflex agents.

Disadvantages

- Requires additional memory
- More complex than reflex agents.

3. Goal based Agent

Definition -

Goal based Agent choose action that achieve a specific goal

Working in smart home

- Ex: Goal - maintain Room Temperature at 24°C
- If temperature increase, switch on the air conditioner
 - If temperature decreases, switch off the AC.

Advantages

Flexible

Disadvantages

Require more computation utility Based Agent.

4. Utility Based Agent

Definition - Utility Based Agent selects the action with highest overall benefits and Balances comfort energy efficiency.

Ex: It balances : user comfort, energy efficiency.

Adjust AC and lighting to maximize comfort while minimizing electricity usage

Advantages -

Makes optimal decision with multiple objectives

Disadvantages -

Utility calculation can be complex.

model based reflex agent.

Conclusion - Goal based Agent, is considered as the best compared to utility Based Agent, simple reflex Agent.

Robot in an unknown maze using UCS and IDS

Problem Statement

A robot is placed in an unknown maze and must reach the exit without knowing the layout.

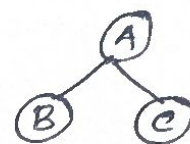
The robot should explore the maze and find a path.

Uniform cost search (UCS)

uniform cost search expands the node within the lowest cumulative path cost.

Ex :

Paths -



$A \rightarrow B \rightarrow \text{Goal} = \text{cost } 10$

$A \rightarrow C \rightarrow \text{Goal} = \text{cost } 6$

UCS selects the second path because its total cost is lower.

Advantages

complete path

Finds the least-cost path

Suitable when movement cost vary.

Disadvantages

High memory usage

slow for large mazes

Complexity

Time complexity - $O(b^{(1+C/\epsilon)})$

Space complexity - $O(b^{(1+C/\epsilon)})$

where,

b = branching factor

C^* = optimal path cost

ϵ = minimum step cost

Iterative Deepening search (IDS)

Iterative Deepening search (IDS) repeatedly performs Depth-limited search with increasing depth limits.

Ex:
Depth 0
Depth 1
Depth 2

Continue until goal is found

Advantages

low memory usage

complete path

Suitable when solution depth is unknown.

Disadvantages

Repeat node expansion

Not optimal if step cost differ.

complexity

Time complexity - $O(b^d)$

Space complexity - $O(bd)$

where,

b - branching factor

d - depth of solution.

Conclusion - when compared to the both uniform cost search (UCS) and Iterative Deepening search (IDS) have complete and suitable when solution depth is cost.

Goal based + Iterative Deepening search is recommended as the best algorithm for path findings.